



NETWORK INSTALLATION PROPOSAL

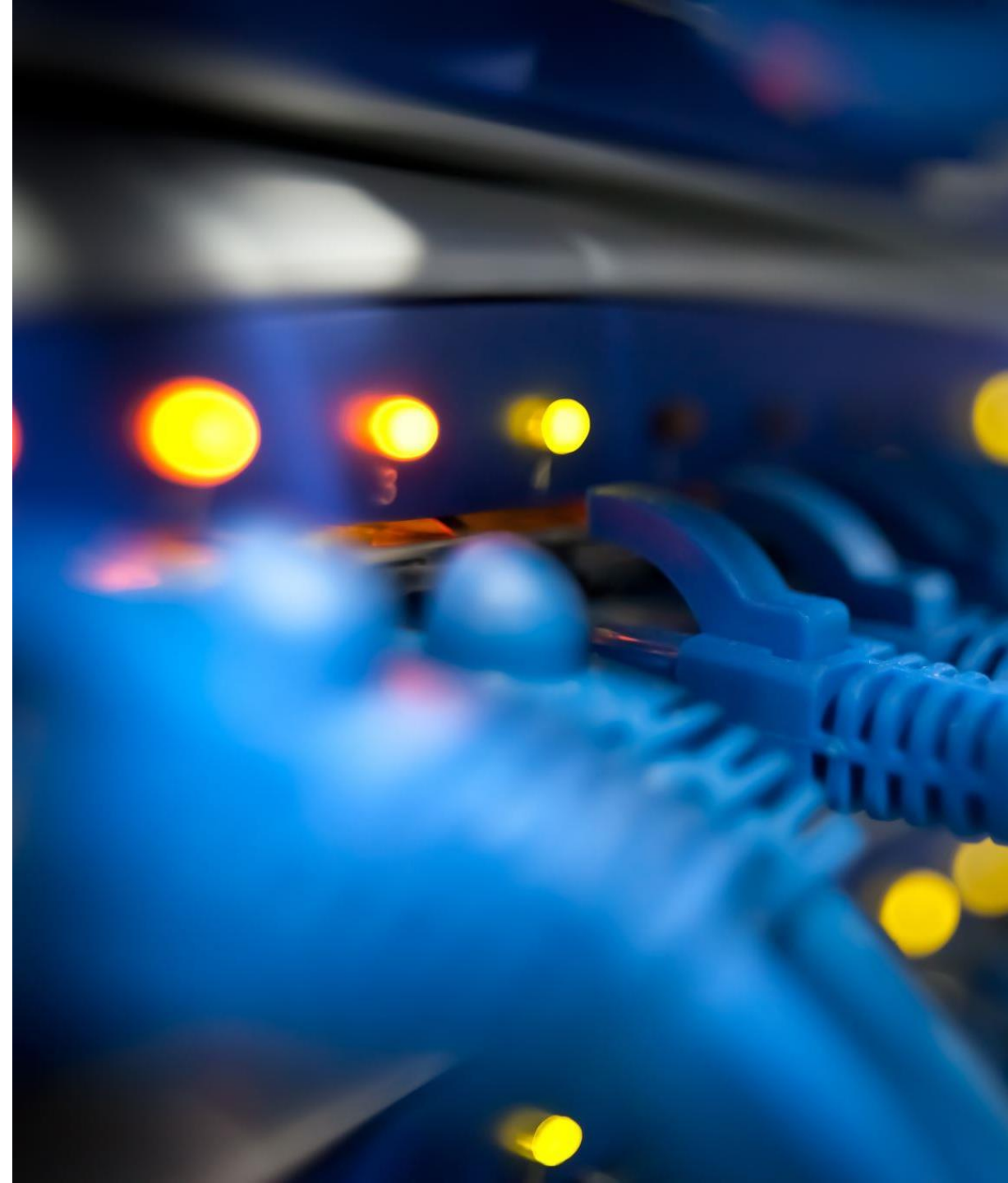
Akshar Patel

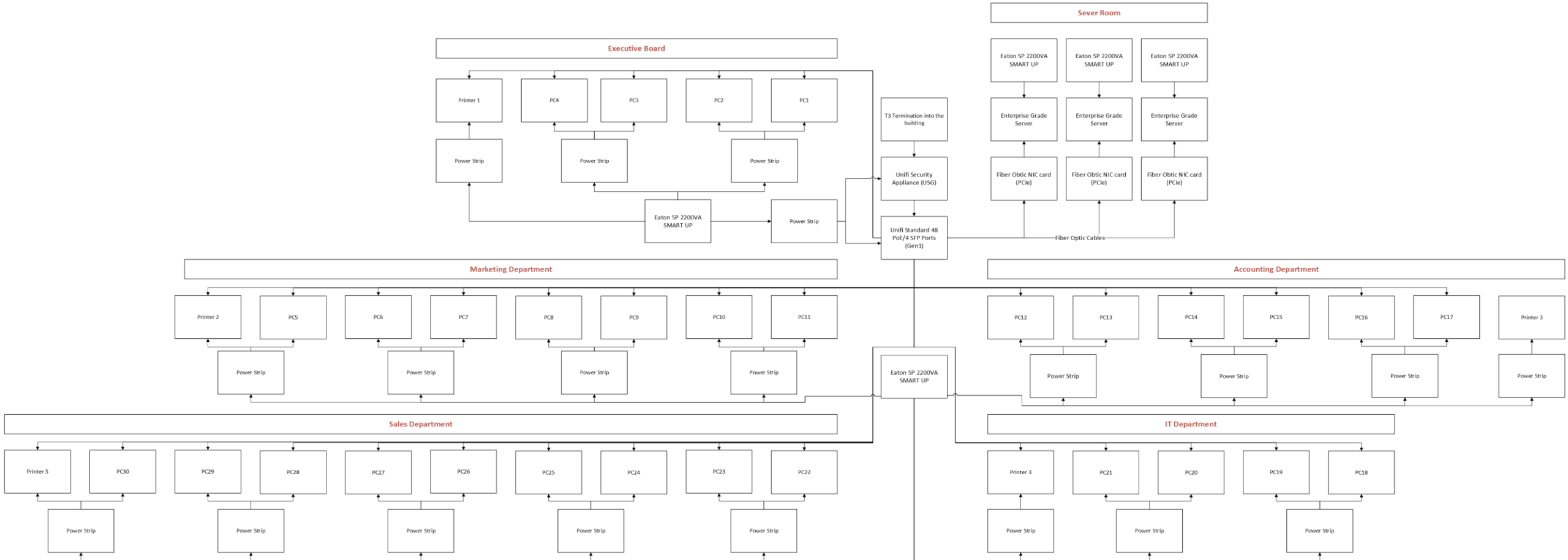
4/21/2025

CEG 2400

NETWORK SUMMARY

- This Network has the capacity to handle 30 PCs; network connected printers and multiple servers and still give you reliable speeds.
- Everything in this network is connected via internet, PCs, printers and servers. (This will allow remote print and remote connection from home)
- This Network will cost under \$100,000, includes purchasing products and installation
- This network insures everyone regardless of position has access to all fundamental aspects of their jobs
- Will allow IT supervision and upmost security for all.





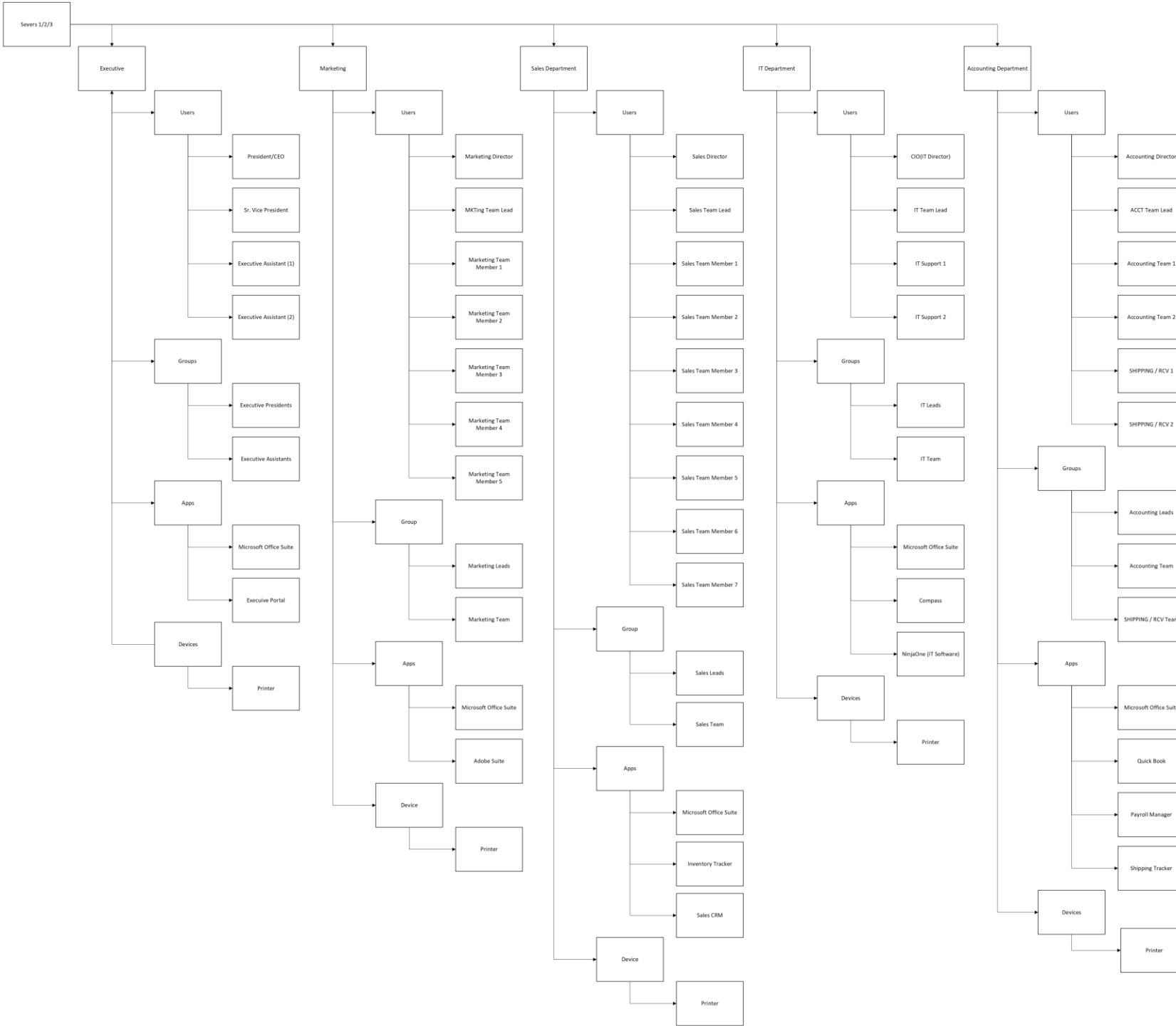
Network Diagram

- This network will connect all employees into servers in which data can be shared and accessed anywhere in the world
- Each department has a dedicated printer (Further explained in the Network Tree/User Diagram)
- There is a SMART UP that connects all 4 department, but this is highly not possible, and connections will most likely come straight out of outlets and only the Executive Board will be connected to SMART UPS
- An expanded version can be found in the folder.

Price List and Parts Summary

Device Name	Quantit	Link	Final Pric	Details
Unifi Standard 48 PoE/4 SFP Ports (Gen1)	1	Unifi Standard 48 PoE/4 SFP Ports (Gen1)	\$835	Main Switch
Duplex Fiber Optic Patch Cable(6m)	3	Duplex Fiber Optic Patch Cable(6m)	\$57	Connecting necessary devices with high speed
Fiber Obtic NIC cards (PCIe)	3	Fiber Optic NIC card	\$840	Will let computers to be directly connected to fiber
CAT7, Unshielded Twisted Pair 1000ft	1	CAT7, Unshielded Twisted Pair 1000ft	\$400	Will give us the ability to make all sizes of connections without having extra
Unifi Security Appliance (USG)	1	Unifi Security Appliance (USG)	\$212	Firewall
Enterprise Grade Servers	3	Enterprise Grade Servers	1,080	Enterprise Proliant DL380 G9 Server 2X E5-2660v3 20 Cores 192GB P440 8X 600GB
LG 34" Curved Ultrawide Monitor	30	LG 34" Curved Ultra Wide Monitor	\$7,500	Monitor
Desktop CPUs	30	Desktop CPU	\$45,000	Y60 Gaming Desktop - AMD Ryzen 7 7700X - 32GB DDR5 RAM - NVIDIA GeForce RTX 4060Ti 8GB - 2TB NVMe SSD - Black
Logitech Mouse/Keyboard combo	30	Logitech Mouse/Keyboard Combo	\$6,000	Logitech MX Keys Combo for Business Gen 2, Full Size Wireless Keyboard and Wireless Mouse, with Keyboard Palm Rest, Bluetooth, Logi Bolt, Quiet Clicks,
Hp LaserJet M507X	5	Hp LaserJet M507X	\$6,300	Wireless print, 2-sided printing, Print(B&W)
Power Strip	35	Power Strips	\$455	Surge Protector Power Strip - 8 Outlets with 4 USB (2 USB C) Charging Ports, Multi Plug Outlet Extender, 5Ft Braided Extension Cord
Eaton 5P 2200VA SMART UPS	5	Eaton 5P 2200VA SMART UPS	\$10,325	Eaton 9PX 2200VA Double Conversion Online UPS, Tower/Rack Mountable, 2000W, 10 Outlets (8 C13/2 C19), Energy Star, Zero Transfer Time, RoHS Compliant - 9PX2200GRT (2 Year Warranty)
TOTAL QUANTITY	147	TOTAL PRICE	\$79,004	TOTAL

- The total for the network will cost around 80k
- Keep in mind that an additional 5k allocation will be needed for CAT7 ethernet cord installation.
- An expanded version can be found in the folder.



Network Tree/User Diagram

- Hierarchical Structure, left to right.
- Each department will have its own dedicated printer, but every department will not have dedicated servers, all data will be stored collectively on all 3 servers, but employees will only have access to certain data if they have the correct permissions.
- Each department has a set of apps which only Users and Groups in that department will have access to.
- An expanded version can be found in the folder.